

Focused HTE Meta-Learner Analysis

Cross-scenario comparison of 8 selected methods

Generated 2026-06-12 | RCT design | β in {0, 1, 2} | 50-50 Monte-Carlo sims/cell

Methods analysed

- DINA Original
- DINA Cross-Fit
- DINA Gao (LASSO k2)
- DINA Centered
- DINA Centered Expanded
- DR-learner (2-Model 2SL)
- R-learner (LASSO)
- Dandl Extension

Scenarios (CATE structure)

- S1 Step in X11 (probit)
- S2 Linear in X14
- S3 Subgroup AND (X14 & X1)
- S4 Subgroup OR (X14 | X4)

$\beta = 0$ is the null (no heterogeneity); larger β = stronger HTE.

Key findings

Quantitative findings: read them directly from the ranking table on the next page and the numbered metric panels in this folder (focus_overall_ranking.csv, focus_summary_by_cell.csv). A hand-written narrative is intentionally omitted so the report cannot drift out of sync with the underlying data.

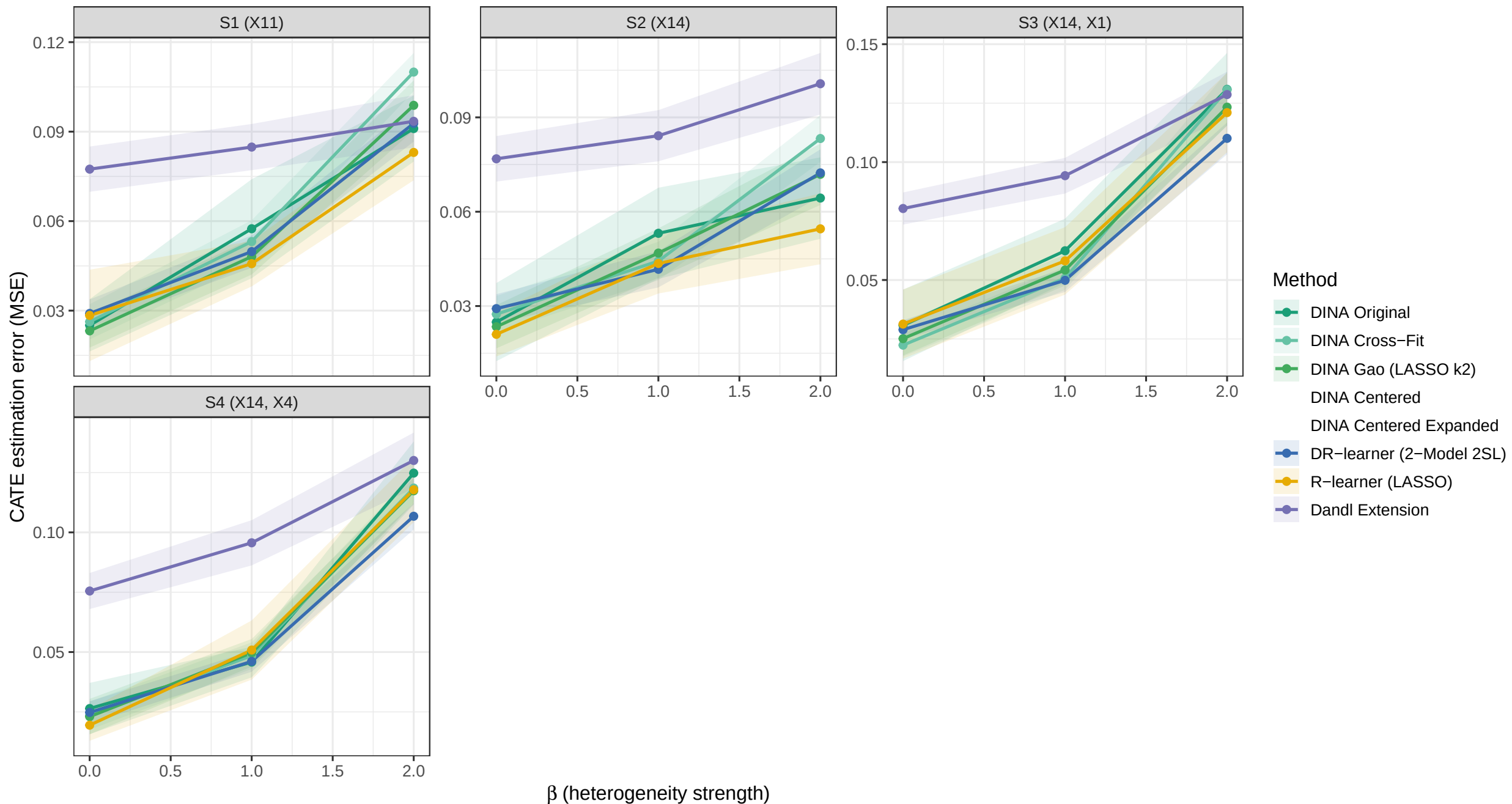
Scenario set (Sun et al. 2022 benchtm): S1 step in X11 (probit), S2 linear in X14, S3 subgroup AND (X14 & X1), S4 subgroup OR (X14 | X4). beta in {0,1,2} (beta=0 = null / type-I error). True effect modifiers per scenario are derived from the DGP in R/scenario_meta.R::get_true_modifiers().

Overall average rank across all scenario x beta cells (1 = best)

Method	Avg MSE rank	Avg correlation rank
R-learner (LASSO)	2.50	1.38
DINA Gao (LASSO k2)	2.75	4.75
DR-learner (2-Model 2SL)	2.75	4.00
DINA Cross-Fit	3.58	6.00
DINA Original	3.75	3.12
Dandl Extension	5.67	1.75

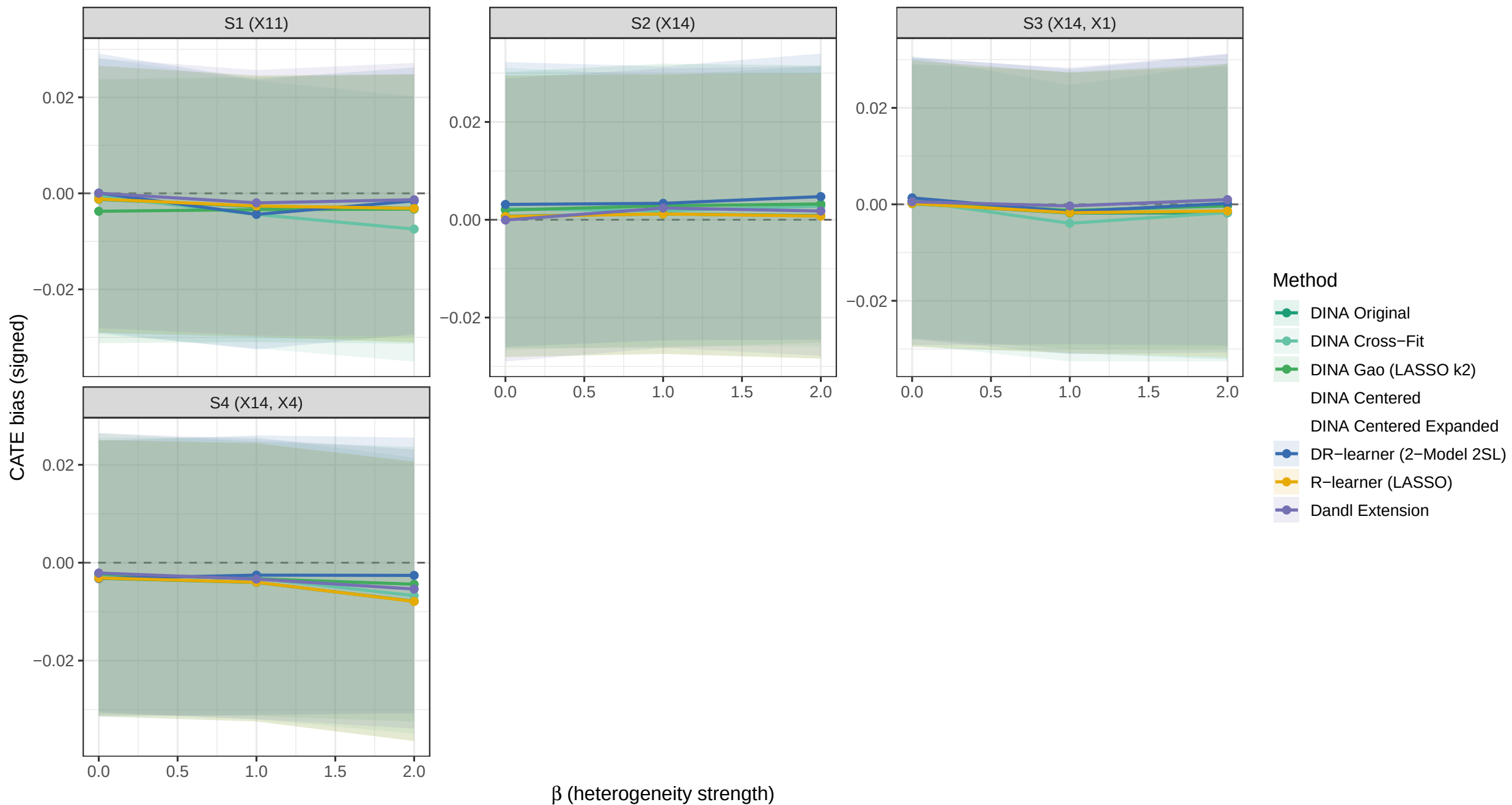
CATE estimation error (MSE)

Lower is better. Ribbons = 95% Monte-Carlo CI. $\beta=0$ is the null (constant effect).



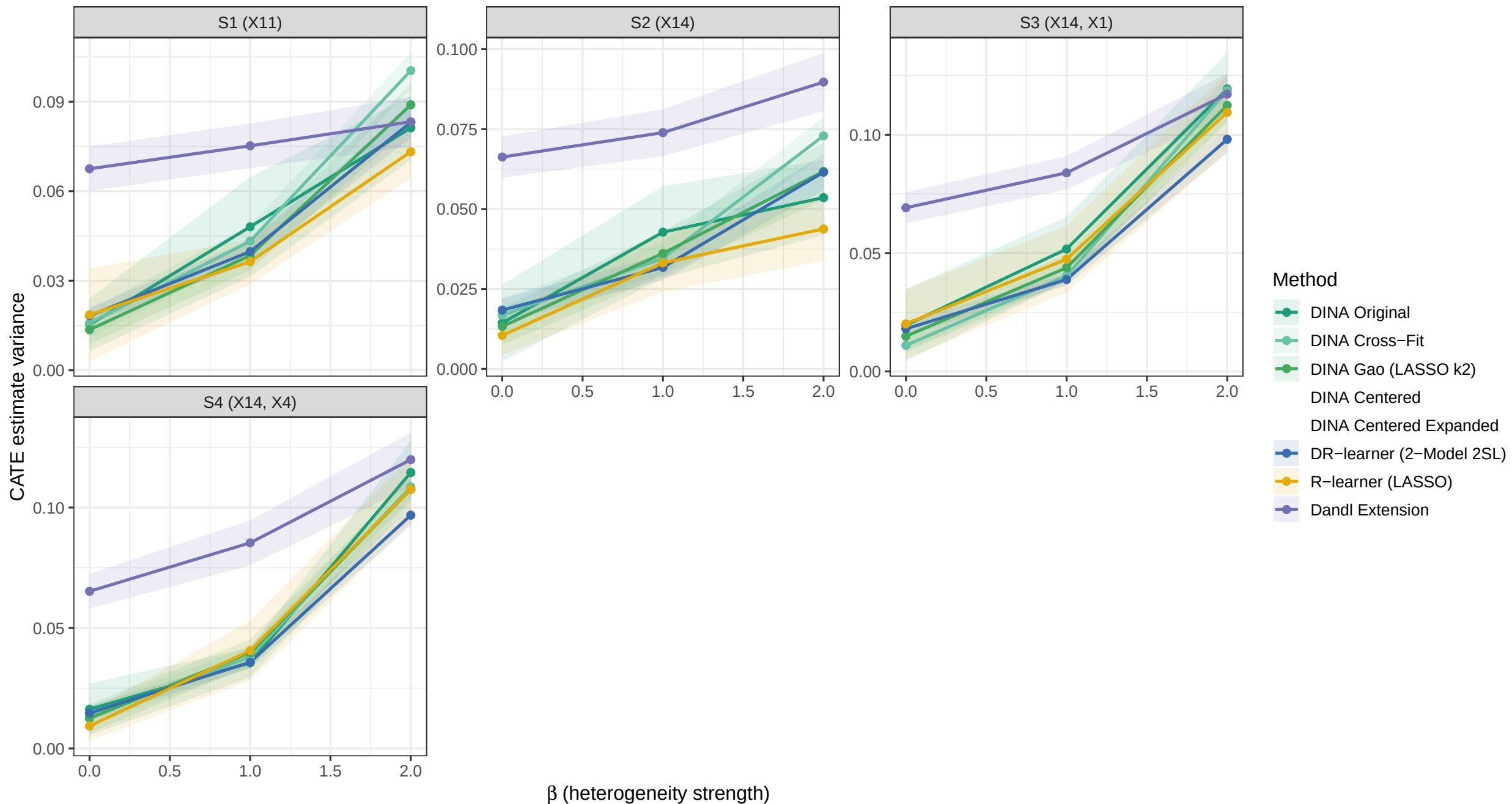
CATE bias (signed)

Closer to 0 is better. Dashed line = 0.



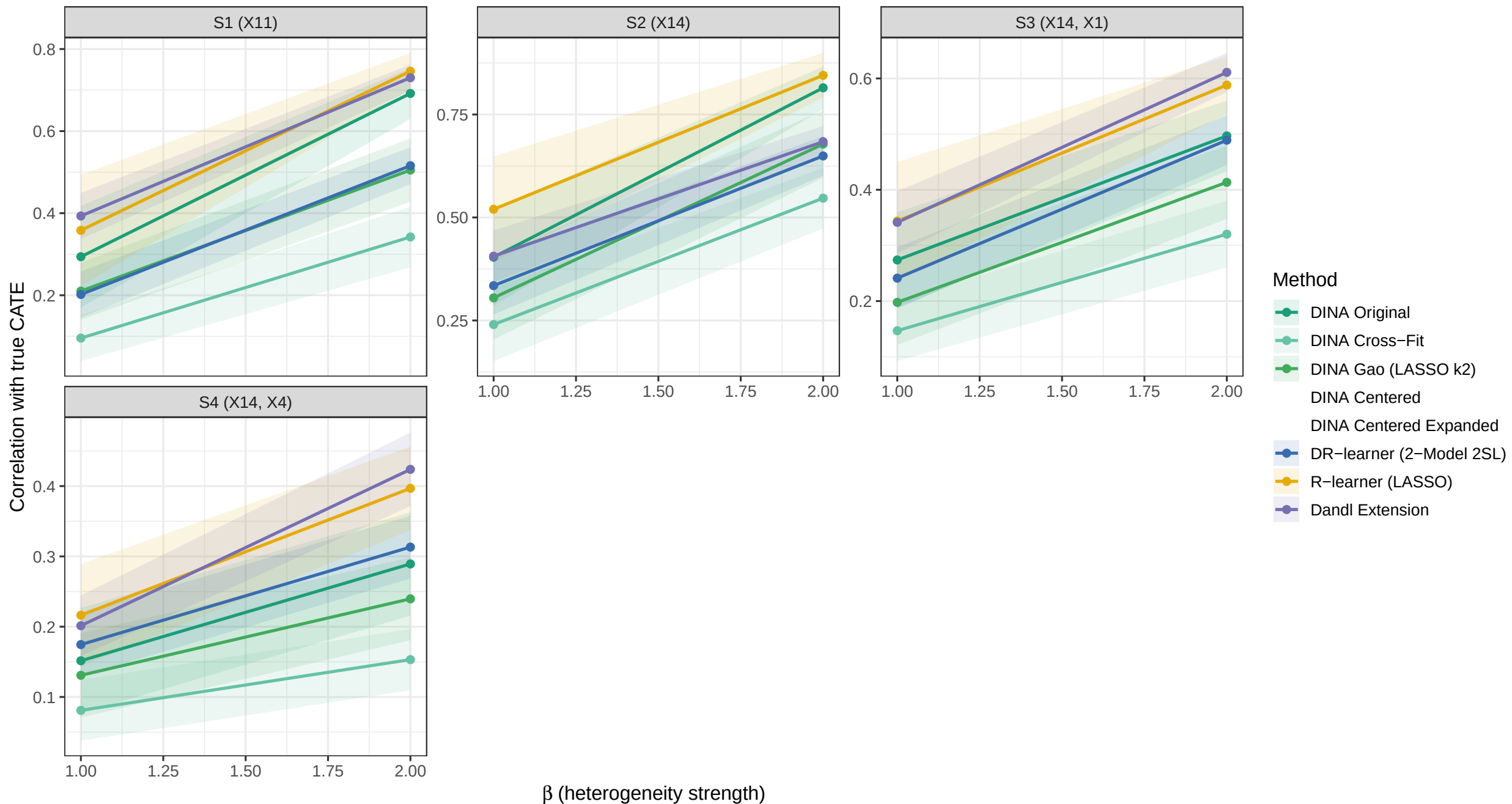
CATE estimate variance

Lower = more stable across the sample. Ribbons = 95% MC CI.



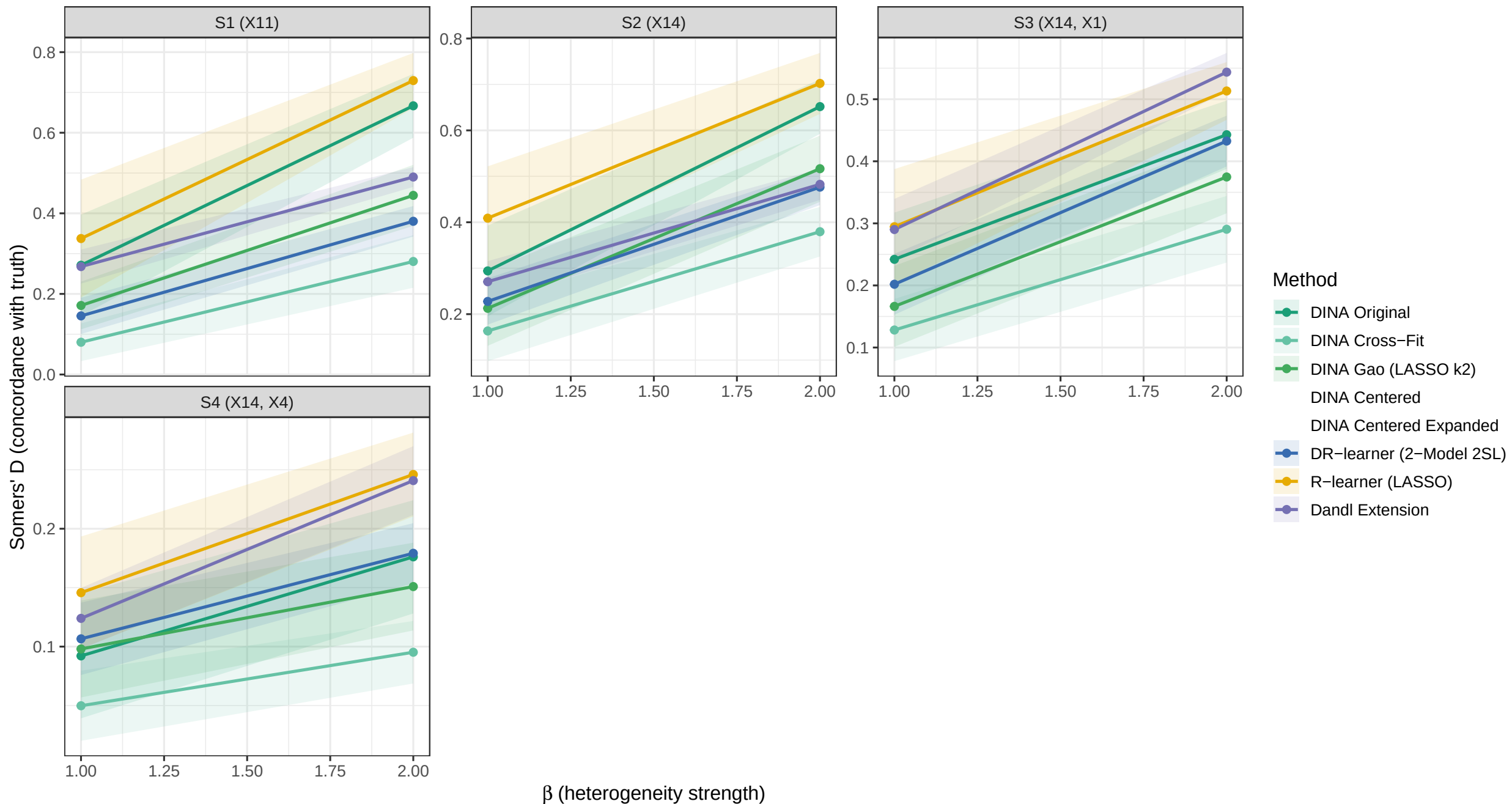
Correlation with true CATE

Higher is better (ranking quality). Undefined at beta=0, so shown for beta>0.



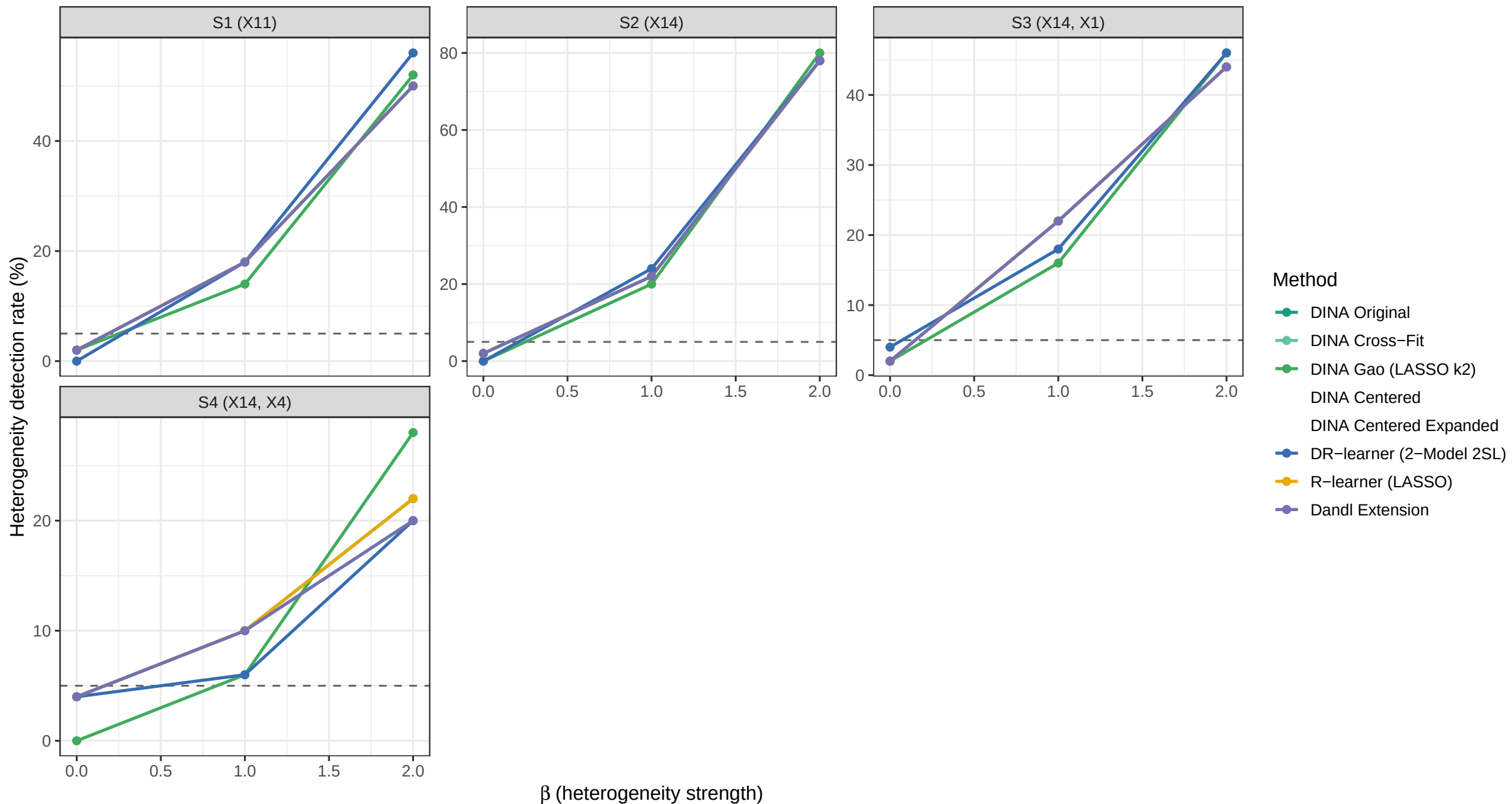
Somers' D (concordance with truth)

Higher is better. Shown for $\beta > 0$.



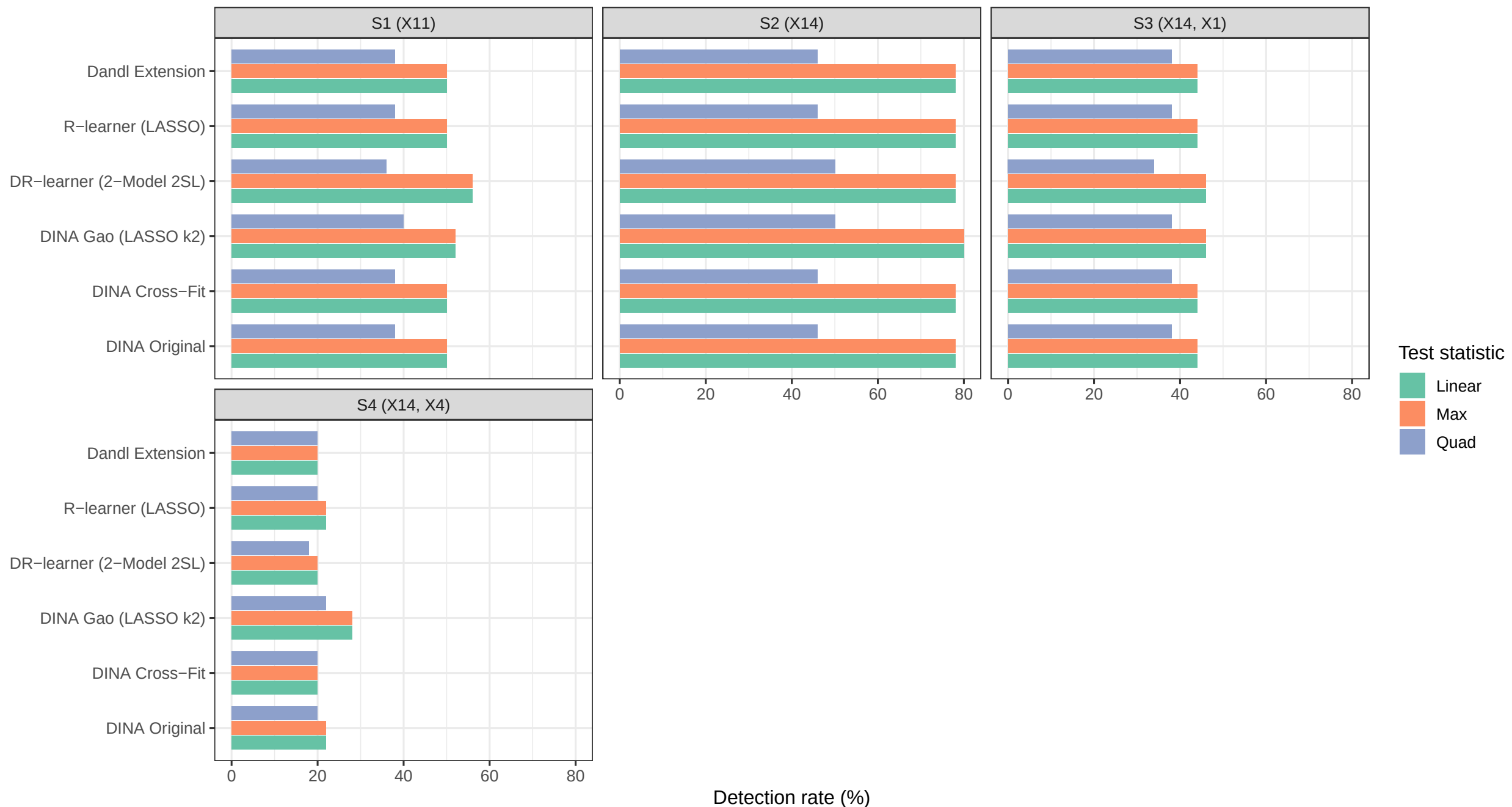
Heterogeneity detection rate (%)

beta=0 = type-I error (target ~5%); beta>0 = power. Dashed line = 5%.



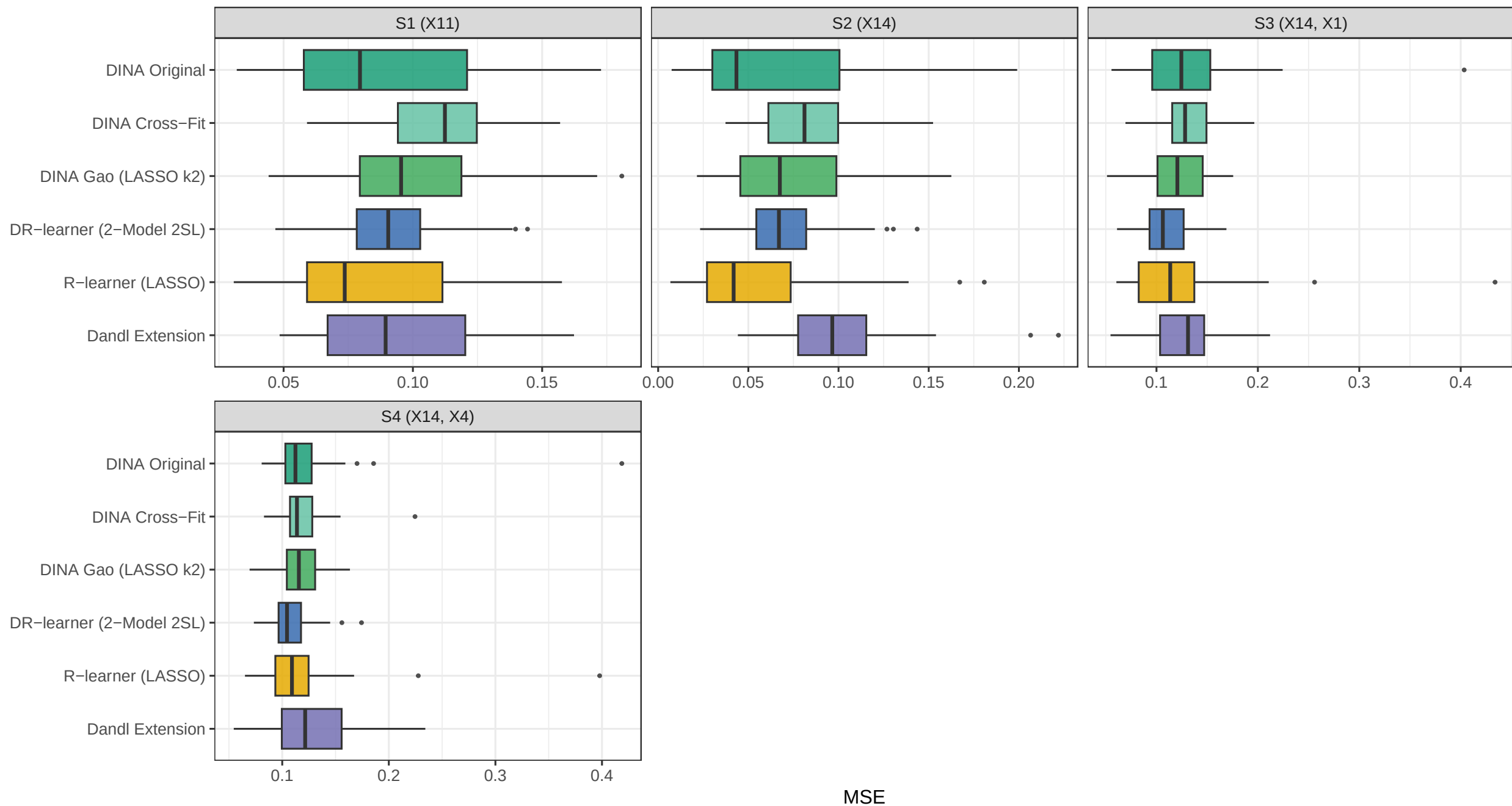
Detection rate by test statistic (beta = 2)

Linear vs Max vs Quadratic heterogeneity test, strongest signal.



Per-simulation MSE distribution (beta = 2)

Spread across the Monte-Carlo replications.



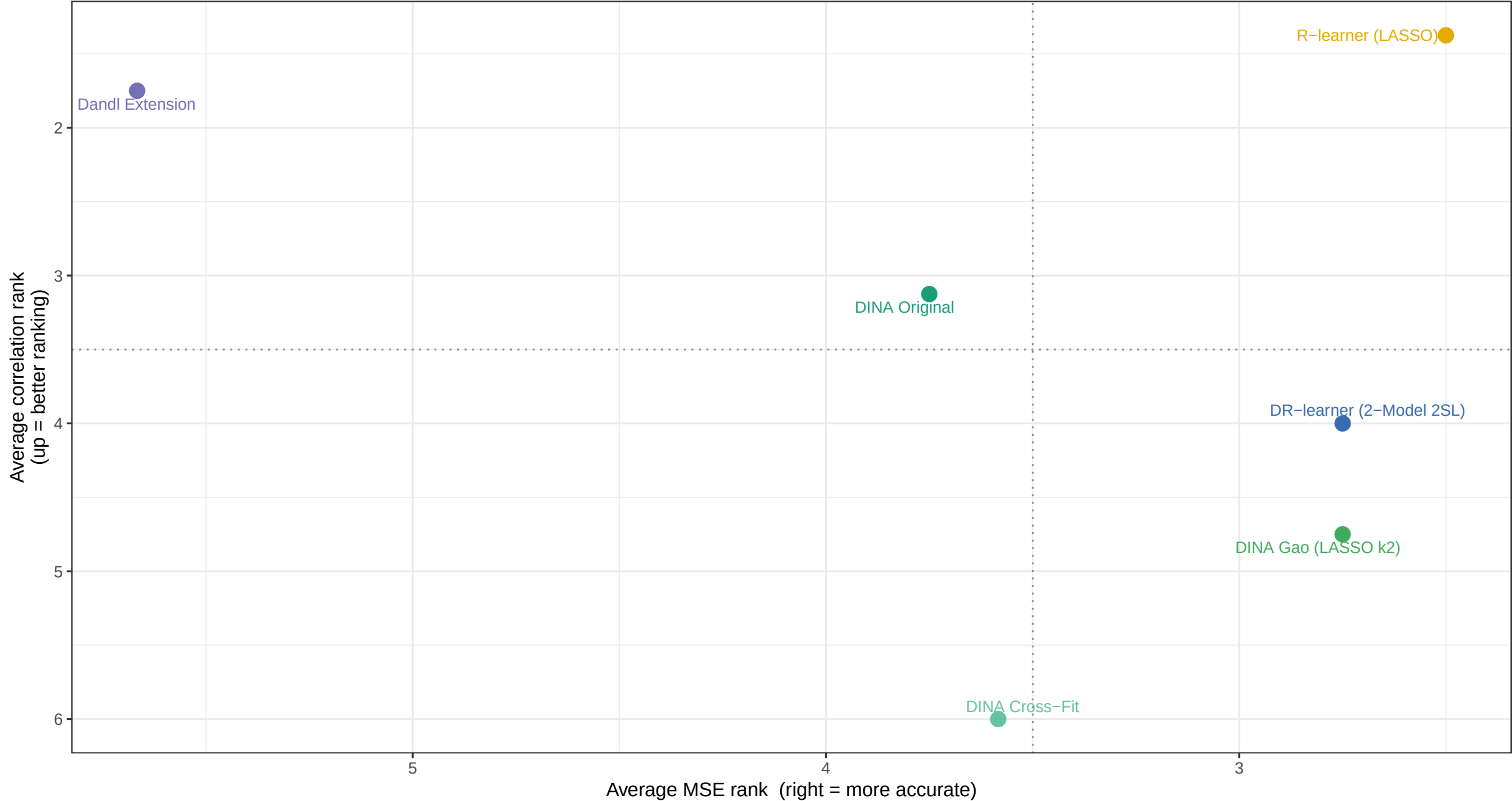
Cross-scenario MSE: rank within each scenario x beta cell

Green = best (lowest MSE), red = worst. Numbers are the mean MSE.



Accuracy vs ranking-quality tradeoff (avg rank across all cells)

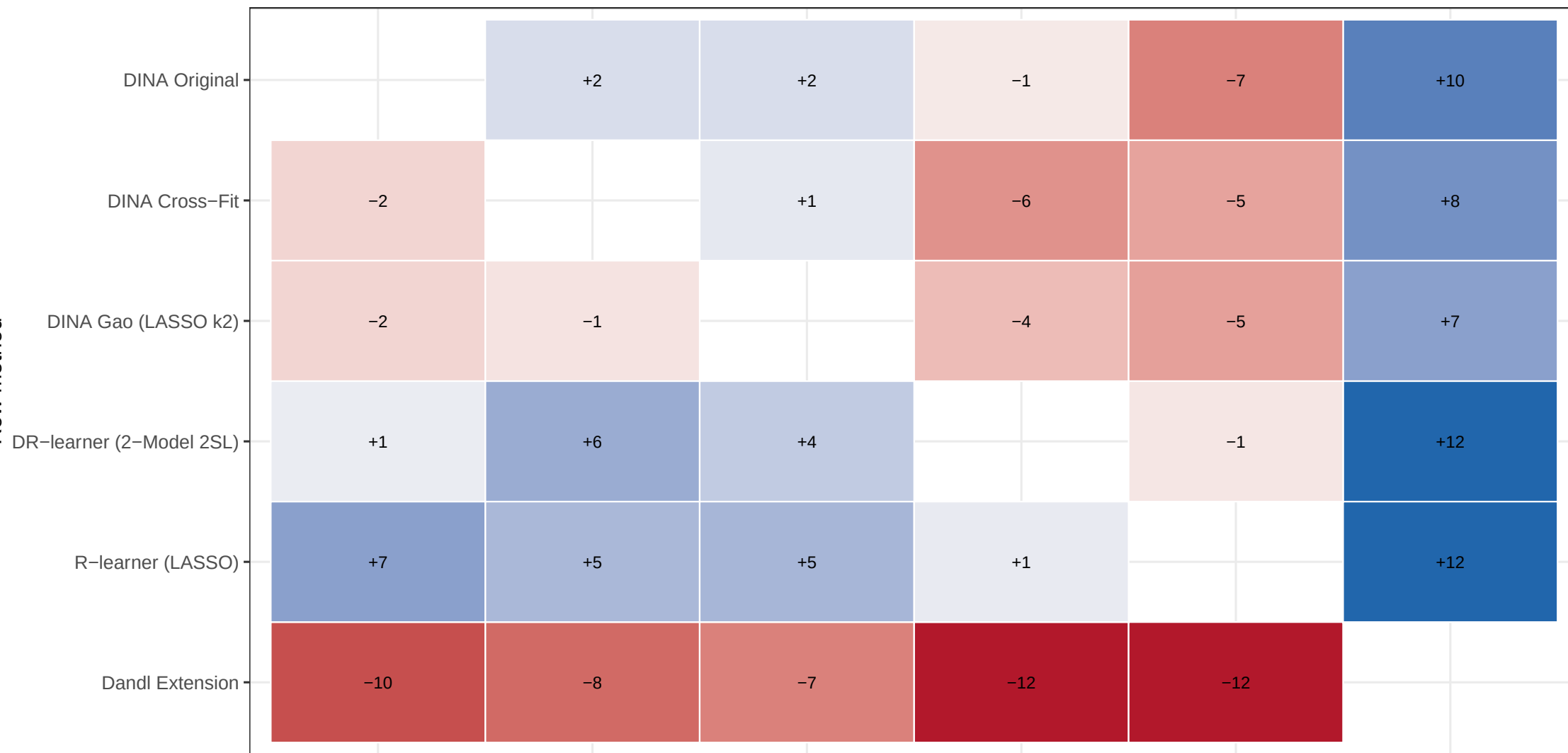
Up & right = better on both. X: MSE magnitude (lower rank better). Y: correlation with true CATE (beta>0).



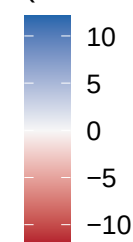
Pairwise head-to-head: net win margin (ROW minus COLUMN)

Avg over all scenarios & beta. Blue (+) = row beats column more often; most pairs tie in ~70–95% of sims.

Row method



Net win %
(row - col)

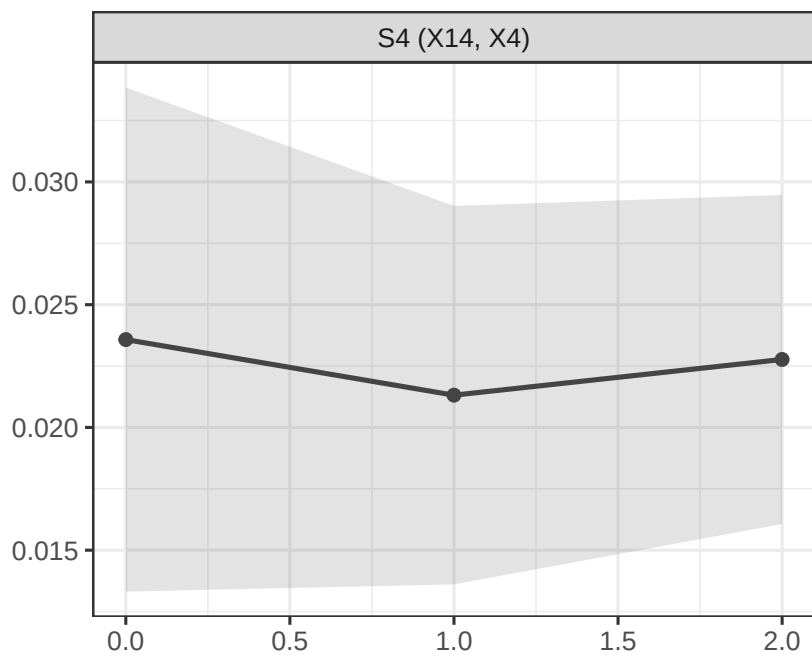
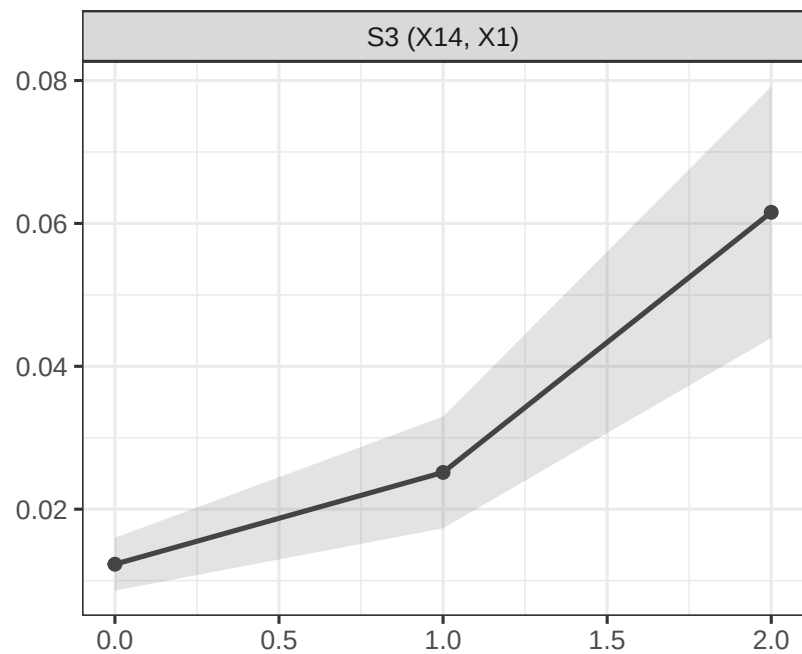
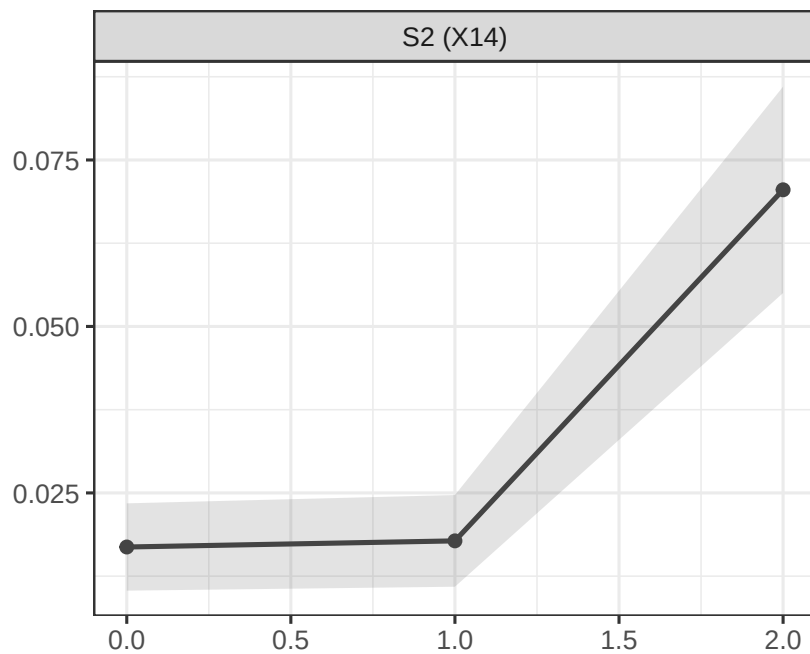
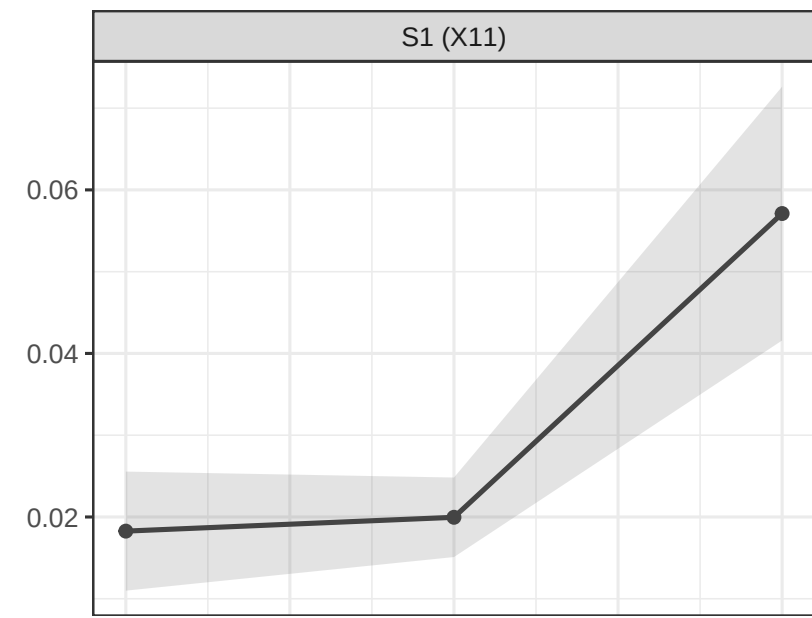


Column method

TMLE (Li 2026 VTE): estimated variance of treatment effect

VTE grows with beta where heterogeneity is real. Ribbon = 95% MC CI.

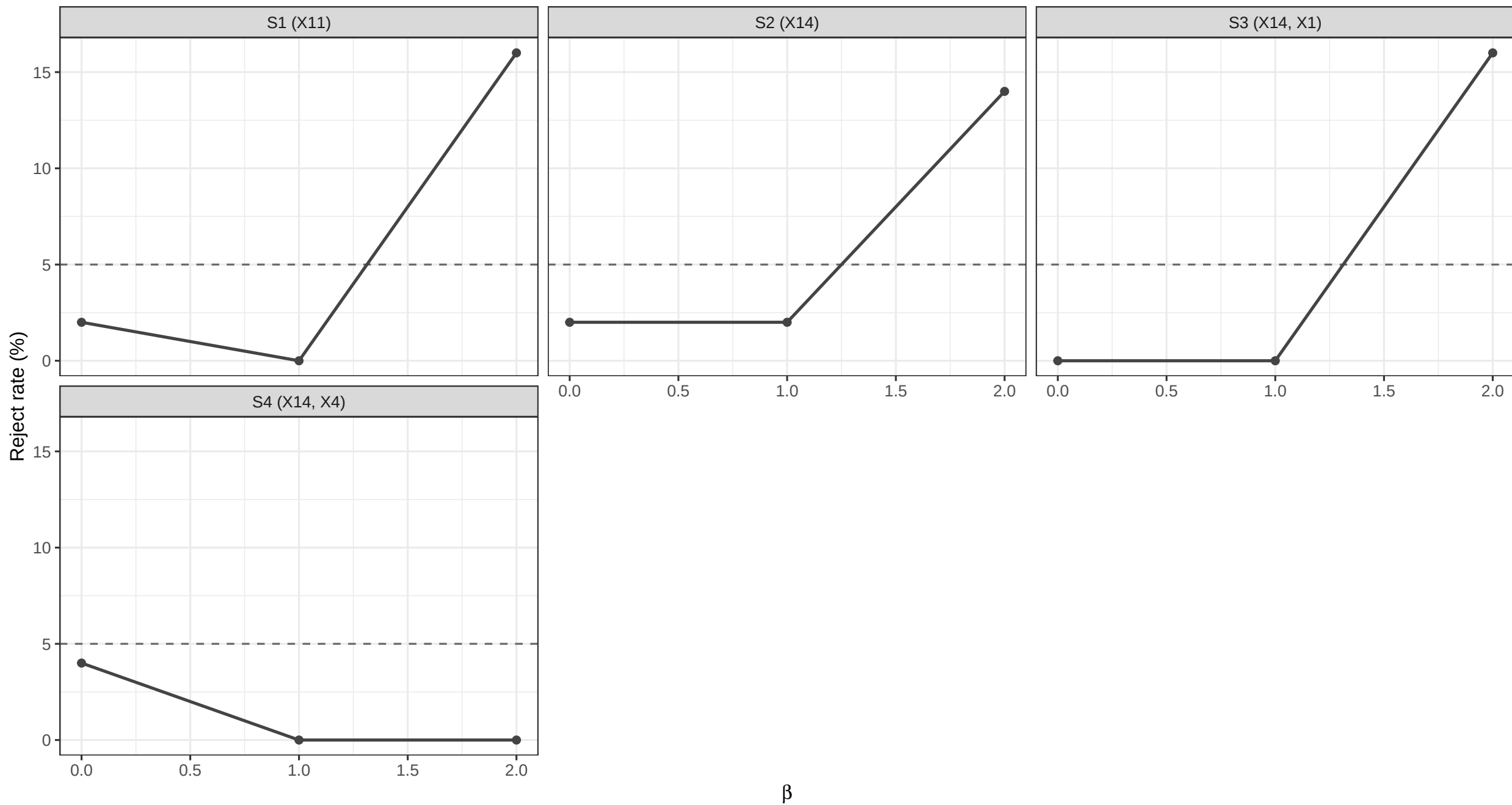
Mean VTE



β

TMLE (Li 2026 VTE): rejection rate (heterogeneity test)

beta=0 = type-I error (target ~5%, dashed); beta>0 = power.



Mean runtime per fit

